

INNOVATIVE MINING SYSTEMS

Mine Back Filling

By

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Mine Back Filling

O/C mine

Dry filling

U/G coal mine

- 1. Hydraulic**
 - Sand
 - Bottom ash
 - Processed O/B
 - Slag
 - Pond ash
- 2. Dry filling**

U/G metal mine

- **Cemented rock filling (CRF)**
- **Hydraulic filling**
- **Cemented hydraulic filling**
- **High concentration filling**
- **Cemented paste backfilling/Paste filling**

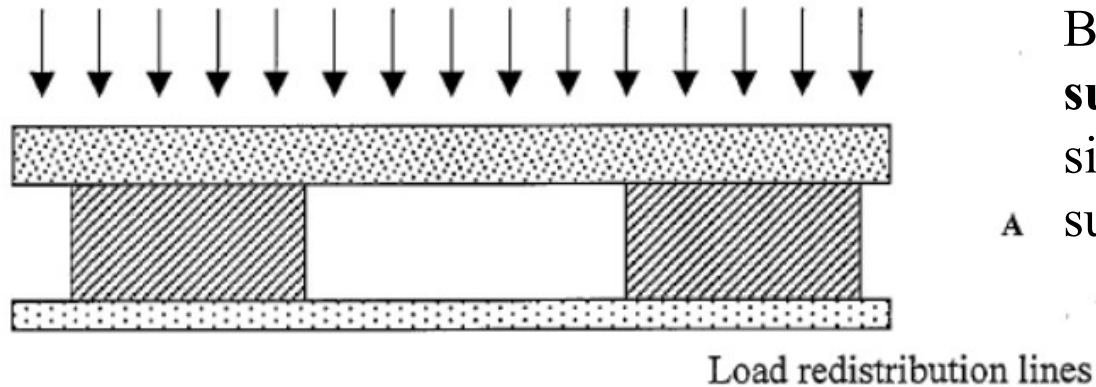
GROUND CONTROL MECHANISM OF BACKFILLING

Backfilling of underground mine workings after mineral extraction has been recognized primarily as one of the important techniques of ground control and is now employed universally in several coal and metalliferous mines not only in India but abroad as a routine work. However, to date there has been no well-established theory clearly expounding the supporting mechanism of the stowed mass or backfill (Zhang et al. 2018a).

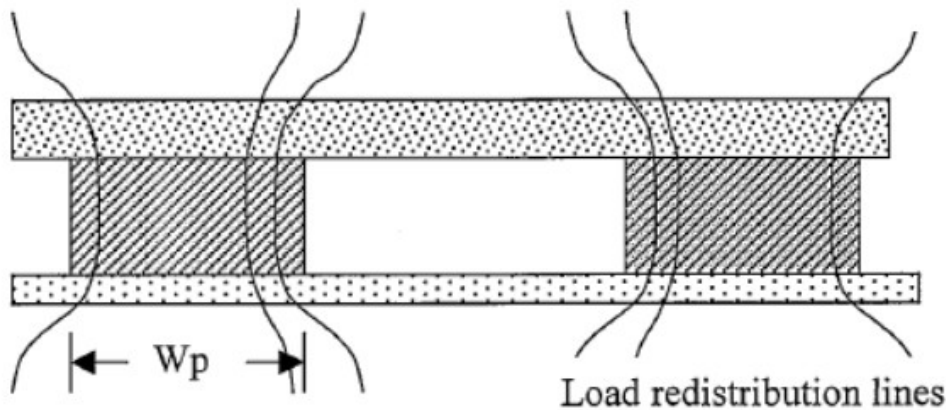
It is generally considered that the backfill supports the rock stress and results into minimal bed separation and sagging of the rock into the opening, and as a result reduces the surface subsidence. It should be remembered that the mechanical properties and packing densities of the fill materials placed in the voids are much smaller than those of the rocks previously existing there before excavation and where fill is subsequently placed. Nevertheless, backfilling, even a partial filling method in coalmines, is widely adopted for ground stabilization (Gupta, 1996 and Behera et al. 2020b).

In metalliferous mines, the ground support function is the most important role for fill as a structural component (Mitchell et al. 1982; Chen and Jiao, 1991; Zou and Nadarajah, 2006; Dirige et al. 2009; Li and Aubertin, 2012; Behera et al. 2023). The presence of fill in any opening can prevent large scale movements and collapse of opening merely by occupying the void. If the fill is sufficiently stiff it can accept some part of the load previously carried by the ore and prevent the load being transferred into the remaining ore. Likewise, stability of the working face can be maintained by maximizing possible ore recovery. Fill acts as a support and can reduce subsidence of workings at higher levels and at the surface, or reduce the load on pillars. The fill as a ground support medium responds by deforming under the imposed back and wall loads.

Load redistribution

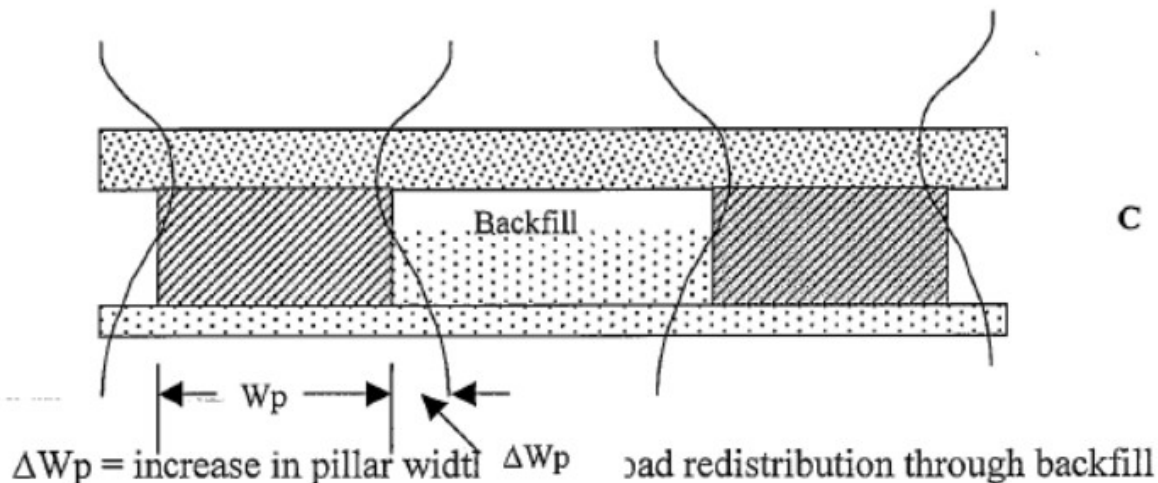


Backfilling is primarily a **passive support**, though in some mining situations it can also provide active support effects.



B

- Passive support acts after rock/ground movement occurs.
- Backfill does not apply a pre-load to the rock mass.
- It becomes effective only when the surrounding rock starts to deform and transfers load onto the backfill.



C

ΔW_p = increase in pillar width ΔW_p

Load redistribution through backfill

Aspect	Active support	Backfilling
Examples	Rock bolts, cable bolts, props	Sand fill, hydraulic fill, paste fill
Pre-loading	Yes	No
Acts before movement	Yes	No
Classification	Active	Passive

Brief history: UG coal stowing

- Past 125-130 years stowing is being practiced throughout the world.
- 1st hydraulic stowing -1864 , Pennsylvania, USA, Idea was given by Catholic Priest to prevent his church.
- The blessing of hydraulic stowing due to flood accidentally discovered in England (Mary Hey Pit).
- 1900-Other European countries.
- 1901-Polland, 1st introduced.
- 1914-France, 1st introduced.
- 1914-More than 100 mines in Germany.
- In India stowing in form of dry manual packing 1st tried in-Mohapani Colliery in Central India (1893)
- Hydraulic sand stowing was first started in Ballarpur Colliery (now in WCL) in March, 1913.
- 1915-Jogta Colliery, Jharia, 1916- Sanctoria Colliery (Raniganj), 1918- Seetalpur II pit, 1919- Bhowra (Jharia), contd..

Brief history: UG metal backfilling

- Mining with hydraulic backfills was introduced in Indian metalliferous mines for the first time in the year 1953 at Kolar Gold Field (KGF), Karnataka. At present around 20 hydraulic backfill plants and 5 paste fill plants are already in operations in metalliferous mines in India.
- The initial backfilling was started with sand and gradually backfilling was shifted to mill tailings. Further upgradation to the backfilling at KGF was done during 1985 (Choudhury and Ghose, 1991). At Mosabani group of mines in Singhbhum District of Bihar (presently in Jharkhand), mining with backfilling was practiced for extraction of chalcopyrite ore bodies dipping at 25-35° (Shetigar, 1983). In continuation with the application of hydraulic backfilling in different hard rock mines, Jaduguda mines, UCIL had also implemented tailings based hydraulic backfilling during 1980s (Choudhury and Ghose 1991).
- In 1977, hydraulic backfilling using tailings was introduced to extract gold deposits at Hutti Gold Mines, Karnataka using overhand cut and fill mining (Thakur et al. 1992). Subsequently, hydraulic backfilling using cemented fill was applied to fill the Large Diameter Bore Hole (LDBH) stopes for maximizing the recovery from the rib pillars.
- Further, backfilling using sand was implemented in Rajpura Dariba mine, HZL, Rajasthan during 1985 and later sand was completely replaced by tailings (Rathore and Kumar, 1992). Post pillar stoping with backfilling was deployed for the first time in Mosabani Group of Copper mines in 1973 and it was also introduced in Rajpura-Dariba mines for extracting thick lead-zinc ore bodies using Vertical Retreat Mining (VRM) and Blast Hole (BH) stopes 1990 onwards (Choudhury and Ghose, 1991).

Mine Back filling

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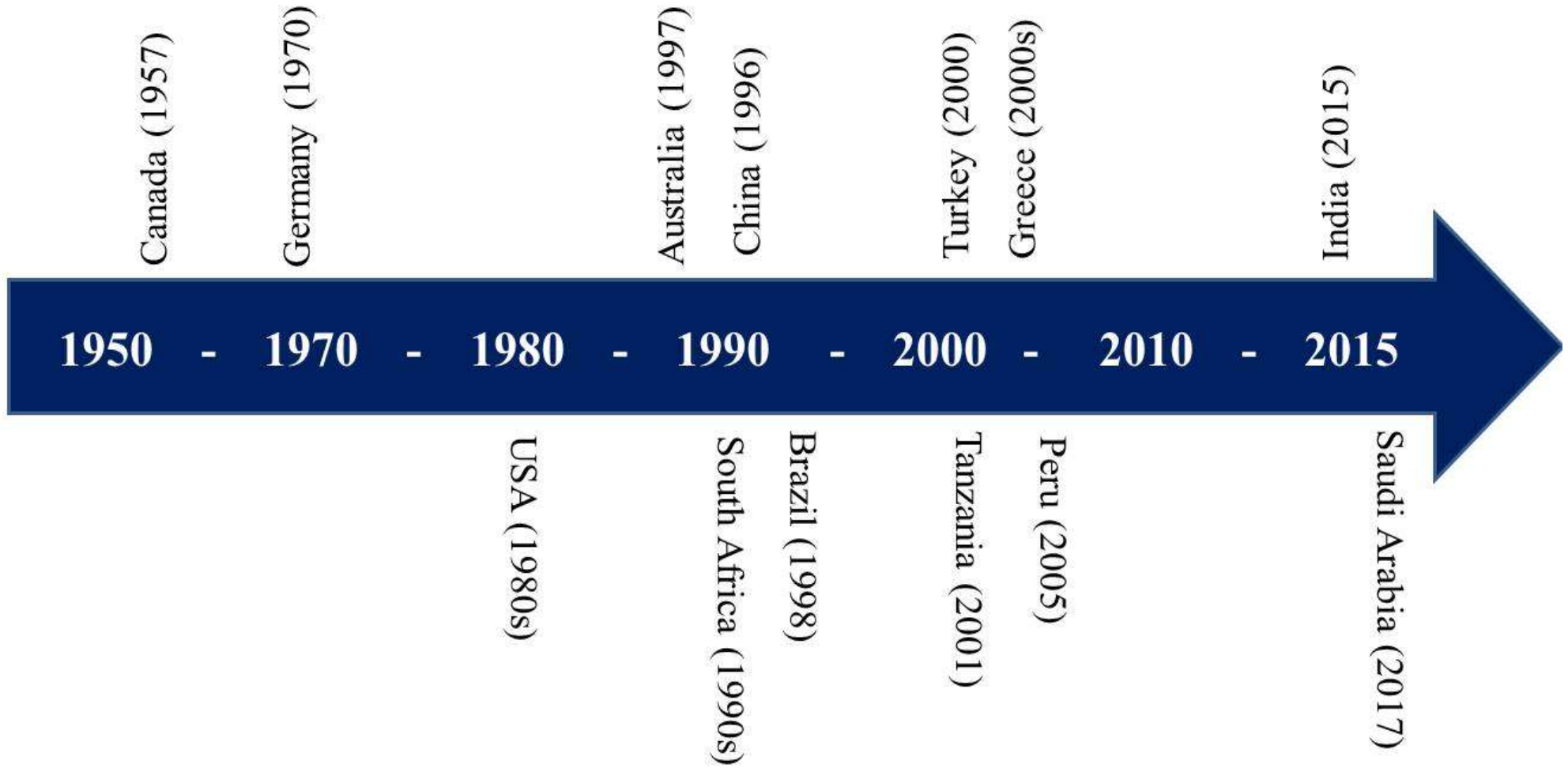
Application of paste backfill: Global scenario



Historical development of paste backfilling

Mine/Country	Year	Reference
Hardy mine, Falconbridge Nickel Mines Ltd., Sudbury, Canada	1957	Tariq and Yanful (2013)
Bad Grund Mine, Germany	1970s	Tariq and Yanful (2013)
Lucky Friday mine, Idaho, USA	1980s	Brackebusch (1994)
South Africa	1990s	Tariq and Yanful (2013)
Jinchuan Group Co., China	1996	Wu et al. (2017a)
Henty Gold mine, Australia	1997	Henderson et al. (1998)
Brazil	1998	Stone (2014)
Greece	2000s	Stone (2014)
Turkey	2000	Stone (2014)
Tanzania	2001	Qi and Fourie (2019a)
Peru	2005	Qi and Fourie (2019a)
SK mine, HZL, India	2015	Behera et al. (2020a)
Saudi Arabia	2017	Brown et al.(2019)

Application of paste backfill: chronology



COMPARISON OF DIFFERENT METHODS OF STOWING IN COAL MINES

Hand Packing

Hand packing, a traditional stowing method in coal mining, where fill materials, such as sand, rock, or other inert materials, are placed manually into mined-out areas. The process involves collecting and transporting fill material, manually placing material into voids and compacting them to provide roof support and prevent collapse, reduce subsidence and ground movement, manage ground pressure and stability and minimize rockfalls and hazards.

Hydraulic Stowing

According to Gupta and Sinha (1967), hydraulic stowing in coal mining offers several benefits, including:

- Reduced fire and gas risks
- Better strata control and subsidence management
- Safer roadways and workplaces
- Elimination of bumps and air blasts
- Enables extraction in challenging areas (thick seams, under water bodies, railways, etc.)
- Higher coal extraction rates
- Flexible working of contiguous seams
- Improved ventilation and reduced gob fire hazards

Overall, hydraulic stowing enhances safety, efficiency, and coal recovery in mining operations. This method of packing has also been used in India to stabilize old workings, where pillars left earlier were too small and galleries too high and wide to stand the stresses of pillar extraction. It has also been used to form sand barriers against a fire area, and for blanketing on the surface to prevent air leakage to a live fire in shallow mines.

Pneumatic Stowing

Pneumatic stowing involves transporting prepared materials like sand, washery rejects, crushed rocks or boiler ash (under 75mm size) through a compressed air stream via a pipeline (up to 300m long) to fill goaf areas. Water jets wet the material before discharge into the goaf. An efficient system can pack 240-300m³ per shift, consuming 85m³ of free air per minute (Karmakar and Ghosh, 1976).

Pneumatic stowing has several advantages, including:

- No wastewater disposal issues
- No impact on roof, floor, or atmosphere
- Suitable for conditions where hydraulic stowing isn't feasible (flat seams, rise workings, or deep mines)
- Can use coarser materials, reducing waste accumulation
- Less pipe damage and easier control

However, it also has drawbacks, such as:

- Higher power consumption and capital investment
- Potential for higher operating costs
- Requires careful material selection (avoiding sand due to excessive air consumption and dust issues)
- Significant strain on mine transport systems due to large material volumes.

Overall, pneumatic stowing offers benefits in specific conditions but requires careful consideration of costs and logistics.

Mechanical Stowing

Mechanical stowing uses machines like belt stowers, impact stowers, and slushers to fill mined-out areas. The process involves transporting material to the face and directing it into the goaf (Karmakar and Ghosh, 1976). It emerged as a solution to the high costs and power consumption of pneumatic stowing systems.

Mechanical stowing offers several benefits, including:

- Lower costs and power consumption
- Ability to handle uneven-sized materials and debris
- Reduced dust generation
- Flexibility in material dampening without clogging issues

However, its adoption is limited due to:

- Strain on mine transport capacity
- Incompatibility with coal-getting shifts and scraper chain conveyors
- Insufficient machine capacity to complete operations within tight timeframes.

These limitations have hindered its widespread popularity.

APPLICABILITY OF STOWING METHODOLOGY IN COAL MINES

To find the applicability of any stowing methodology in a coal mine, for a particular geo-mining condition, the stowing methods need to be assessed based on the following four broad criteria:

The aim of stowing is to sustain the overlying strata load and minimize deformation of remnant pillars. Filling of voids up to the excavated thickness reduces stress on the adjacent pillars with minimum sagging of roof and subsidence at surface. Filling of the voids less than the excavated height will result into deformation of overlying strata and crushing of adjacent remnant pillars due to transfer of load on these standing pillars. However, in actual, the stowed material suffers shrinkage due to compaction and actual thickness decreases with the increase in roof pressure. If shrinkage of filled material is less than the bending moment of overlying strata, minimum cracking or collapse of roof strata will take place and the surface profile will be stable.

Considering these aspects of stowing, the efficiency of stowing called "packing efficiency" i.e. the ratio of the thickness of the stowed material to that of the original one, plays a major role in strata control (Rai, 1971a).

Hydraulic Stowing

Hydraulic stowing is widely used in India to pack the goaf in bord and pillar as well as longwall workings. In principle, sand or crushed rock is transported underground in the form of sand and water mixture through pipes. The material settles behind barricades of bamboo matting or hessian cloth, and the water percolates to a lower level and collects in specially designed sumps.

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Hydraulic stowing in Indian mines was enabled by factors like proximity to sand sources, no adverse water effects on roof and floor conditions, shallow mine depths, and suitable seam inclination, making it a viable method.

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Numerical Problem

The total stowing time can be split into (i) preparatory time, t_p for pipe shifting, barricade erection, etc., (ii) actual stowing time, t_a and (iii) post-stowing time, t_f for cleaning of face machinery, shifting of pipes and machines, etc. (Smith and Khan 2021)

Thus, total stowing time,

$$T = t_p + t_a + t_f = t_p + \left\{ \frac{l.b.h}{Q_s} \right\} + t_f \quad \text{.....2.4}$$

where,

l =length, b =breadth and h =height of the void

Q_s = rate of stowing (volume).

Example:

In an underground coal mine panel, a void of length 35 m, breadth 5.5 m, and average height 3.6 m is to be filled using hydraulic stowing. The stowing pipeline must be shifted several times during the operation, resulting in a preparatory time of 3.5 hours. After completion of stowing, 2.25 hours are required for cleaning machinery and shifting the pipeline. The hydraulic stowing plant operates at a rated capacity of 50 m³/hr, but due to operational inefficiencies and pipe friction losses, the effective stowing rate reduces by 12%.

Additionally, during the stowing operation, 8% of the void volume to be left unfilled due operational reasons.

- a) Calculate the effective stowing rate.
- b) Determine the effective volume that must be stowed.
- c) Calculate the actual stowing time required to fill the void.
- d) Determine the total stowing time for the operation.
- e) If the mine operates two 8-hour stowing shifts per day, calculate how many days will be required to complete the stowing operation.

Answer:

Effective Stowing Rate Q_s

Rated capacity = $50 \text{ m}^3/\text{hr}$

Loss due to inefficiency = 12%

$$Q_s = 50(1 - 0.12)$$

$$Q_s = 50 \times 0.88$$

$$Q_s = 44 \text{ m}^3/\text{hr}$$

Total Void Volume

$$V = l \times b \times h$$

$$V = 35 \times 5.5 \times 3.6$$

$$V = 192.5 \times 3.6$$

$$V = 693 \text{ m}^3$$

Effective Volume to be Stowed

8% remains unfilled.

$$V_s = 693(1 - 0.08)$$

$$V_s = 693 \times 0.92$$

$$V_s = 637.56 \text{ m}^3$$

Actual Stowing Time t_a

$$t_a = \frac{V_s}{Q_s}$$

$$t_a = \frac{637.56}{44}$$

$$t_a = 14.49 \text{ hr}$$

Total Stowing Time T

$$T = t_p + t_a + t_f$$

$$T = 3.5 + 14.49 + 2.25$$

$$T = 20.24 \text{ hr}$$

Number of Days Required

Two shifts per day:

$$2 \times 8 = 16 \text{ hr/day}$$

$$\text{Days} = \frac{20.24}{16}$$

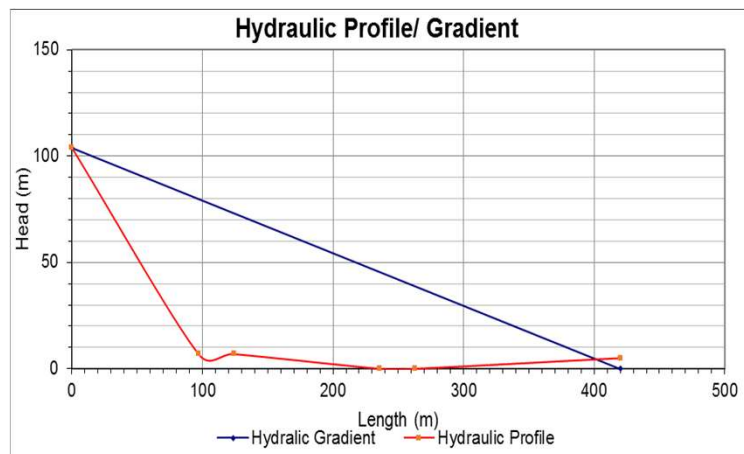
Days = one day and 4 hour of next day

UNDERGROUND PIPE LAYOUT



Pipe Route Alternative

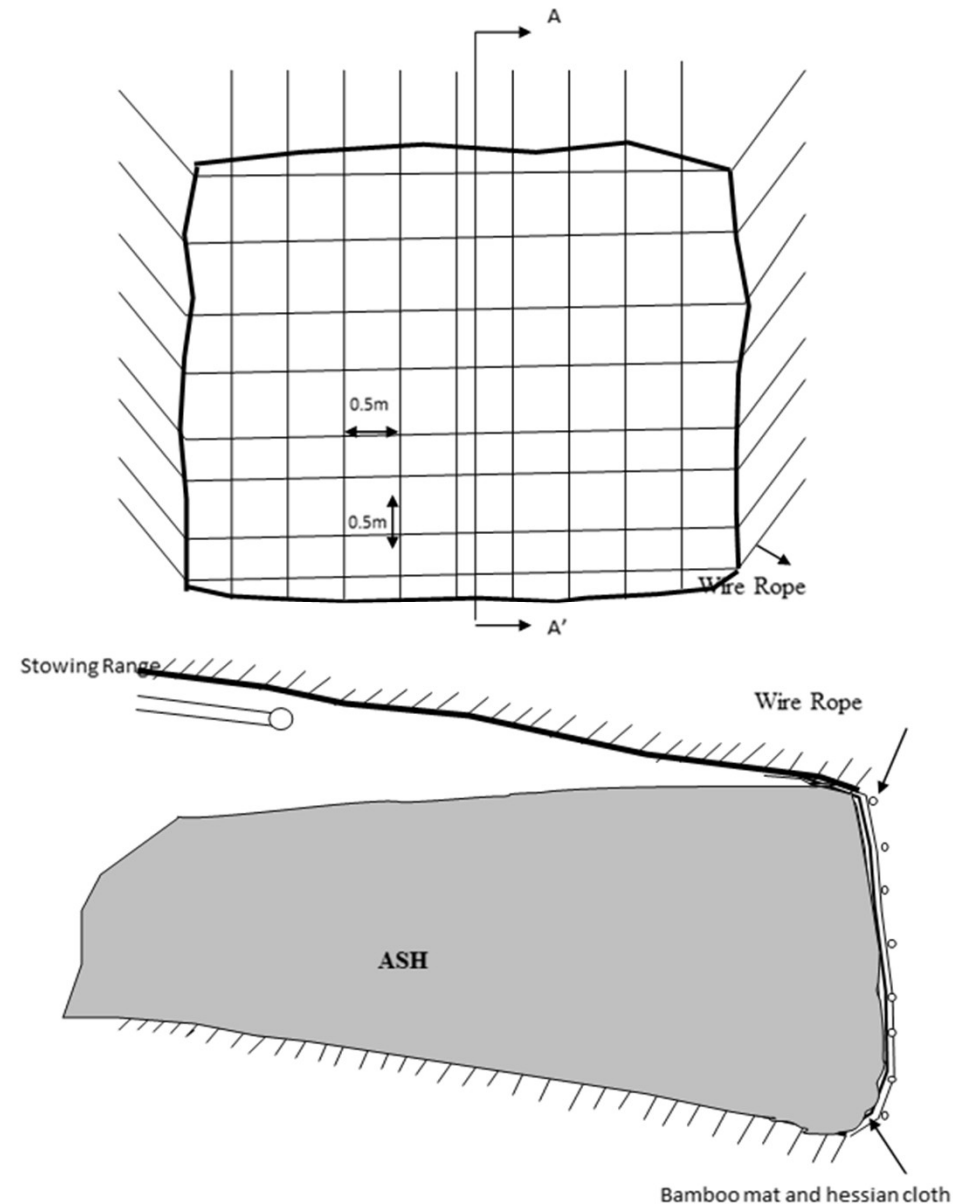
➤ **K'(surface) - K (bottom) -T - U**



Range location	RL	Head diff	Total Head	Pipe Dia.	Total length	L/H
	(m)	(m)	(m)	(mm)	(m)	
BH Top (K')	273.00	0	0	155	0	0.00
5 ED / 50L (K)	176.00	97.00	97.00	155	97	1.00
90 deg. Bend	176.00	0.00	97.00	155	124	1.28
5 ED / 54L (T)	169.10	6.90	103.90	155	235.93	2.27
90 deg. Bend	169.10	0.00	103.90	155	262.93	2.53
11 ED/ 54L (U)	174.10	-5.00	98.90	155	420.07	4.25

DESIGN OF BARRICADING SYSTEM

- Single layer bamboo matting, secured to the sidewalls, roof and floor by wire ropes and wooden cogs.
- Spent wire ropes of 2 – 3 strands.
- Skin-to-skin wooden cogs for providing additional support.
- 0.2% < 53 microns, use additional layer of hessian cloth along with bamboo matting.
- First layer of hessian clothes attached to the second layer of bamboo matting and which in turn rests on the wire mesh.



Bottom ash stowing plant

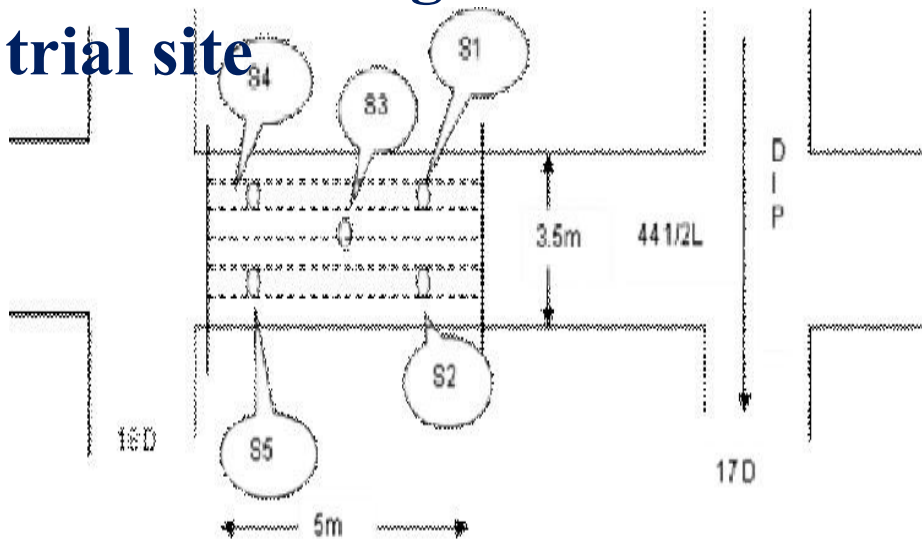






Stowing performance

Schematic diagram of position of monitoring station at the trial site

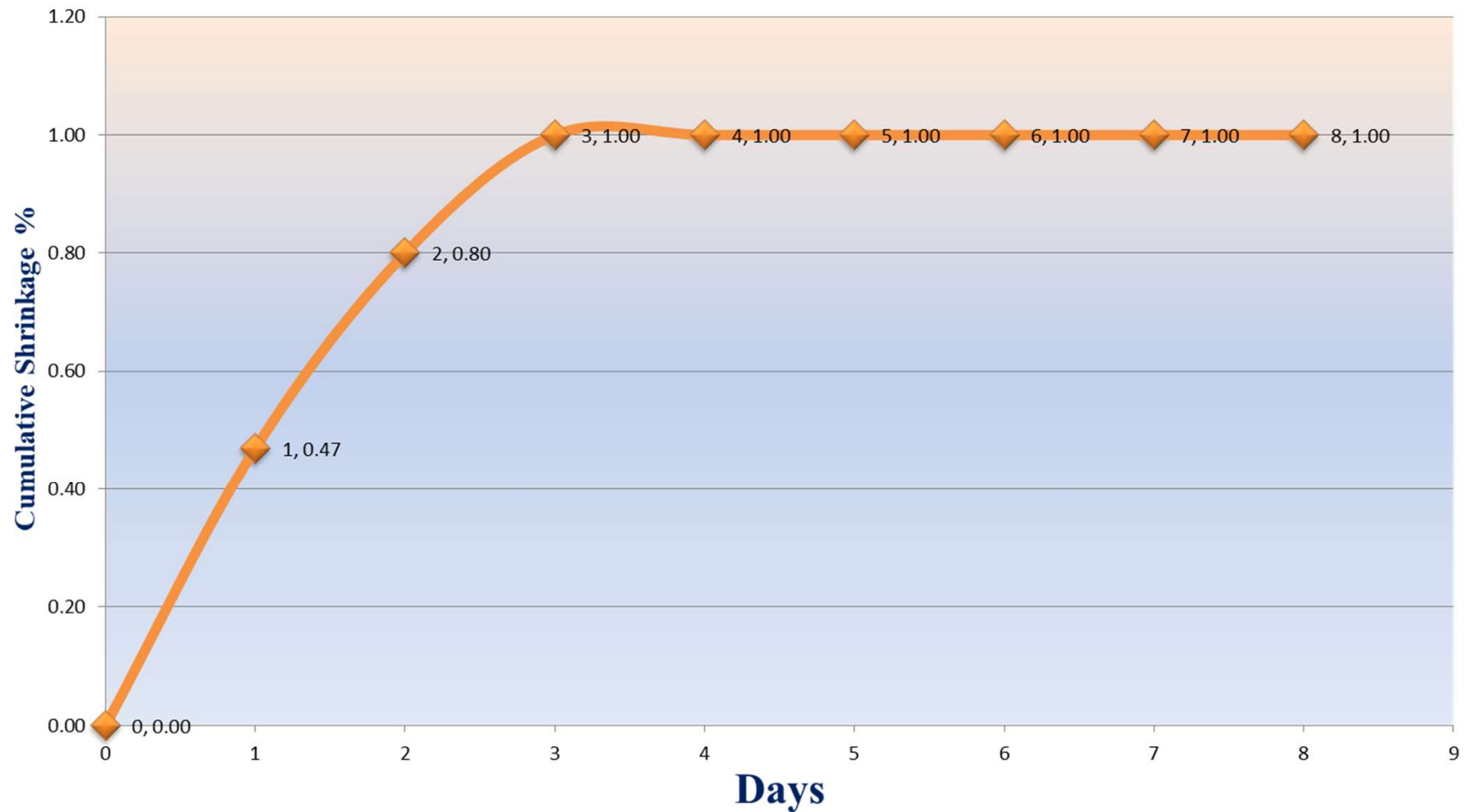


Arrangements before field trial



Barricading arrangement for bottom ash stowing

Shrinkage



All the five stations encountered shrinkage of 15mm during initial 3 days reading.

Stowing performance

- ❖ Stowing rate of 50 – 60 m³/hr was achieved at hydraulic gradient of 1 in 4.3.
- ❖ The fill had fully consolidated.
- ❖ No difficulty for SDL to move on stowed bottom ash pack in the 2nd lift slice.
- ❖ Stowed ash was fully packed and touched the immediate roof.
- ❖ No instances of barricade failure and no puncturing of the rib from adjacent slice.

Self standing height and full packing of bottom ash in the stowed slice



METAL MINE BACKFILLING

1. Rock Filling
2. Hydraulic Slurry Backfilling
3. Cemented Hydraulic Slurry Backfilling
4. High Concentration Filling
5. Cemented Rock Filling
6. Cemented Aggregate Backfilling
7. Paste Backfilling

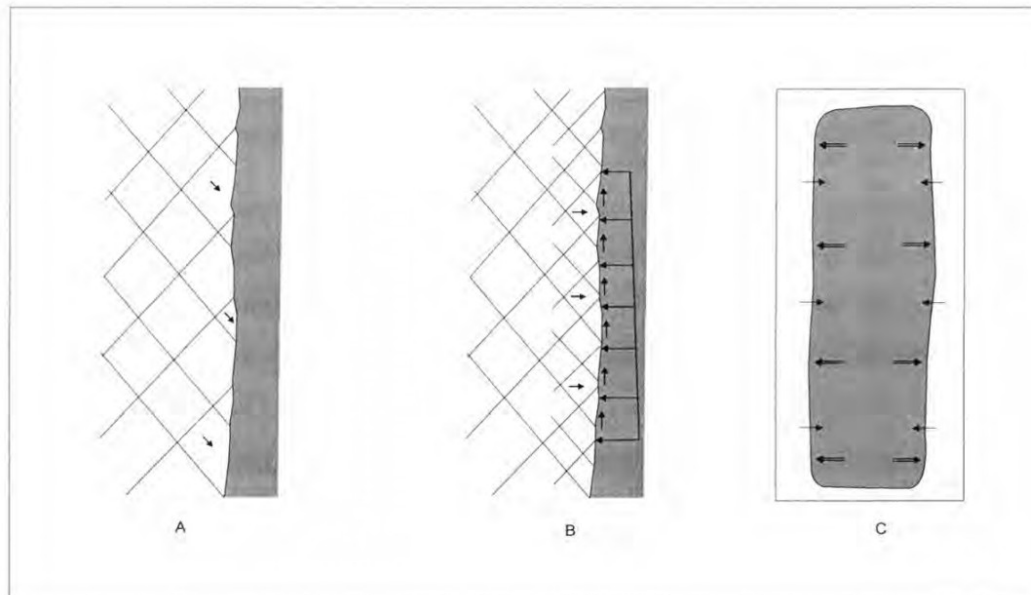
Purpose of backfill (ug metal)

- Backfill is used as a passive support medium.
- In overhand and underhand cut and fill and sub level stoping, backfill is introduced progressively after each ore drive to provide immediate passive support to the hangingwall and footwall
- while backfill placed in open stoping operations is delayed until after the complete extraction of each stope to provide passive support to the hangingwall and footwall.
- The role of backfill as suggested by Barrett et al. (1978) is not to transmit the rock stresses, however to reduce the relaxation of the rock mass so that the rock itself will retain a load bearing capacity.

(Progressive muscle relaxation is a method that helps relieve that tension. In progressive muscle relaxation, you tense a group of muscles as you breathe in, and you relax them as you breathe out. You work on your muscle groups in a certain order. When your body is physically relaxed, you cannot feel anxious)

Purpose of backfill (ug metal)

- Brady and Brown (1993) however propose three main mechanisms that demonstrate the support potential of mine backfill.
- The main mode of support is to provide kinematic constraint to surface blocks in low stressed areas.
- The second mode is the passive resistance of stope wall displacements that can cause high local stresses within the stope wall.
- The third mode proposed by Brady and Brown is the global support provided when the fill mass is confined. This support mechanism is generally only offered by backfill in steeply dipping narrow vein mines.



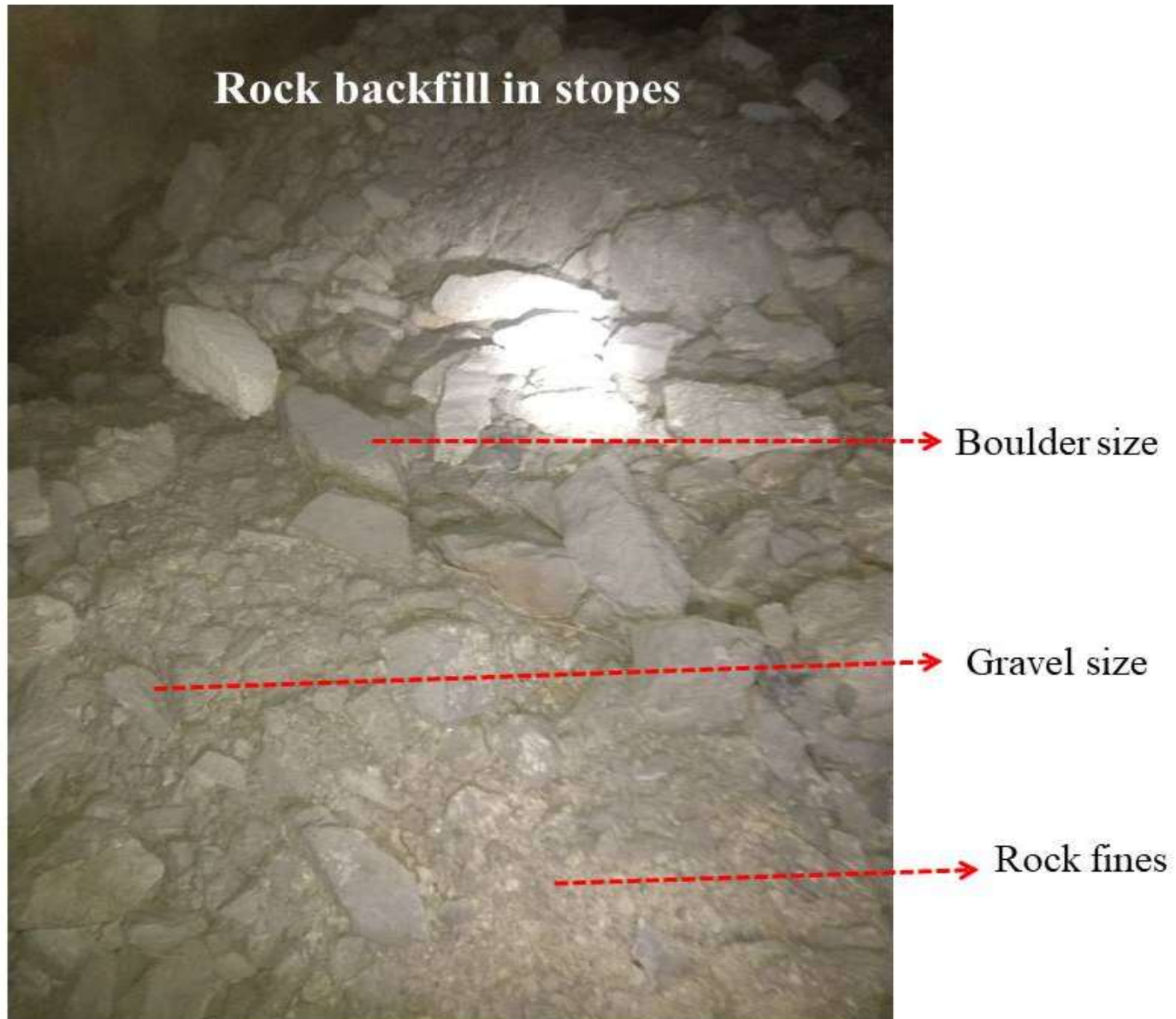
METHODS OF BACKFILLING IN METALLIFEROUS MINES

Rock Filling

A huge amount of waste rock is generated during the extraction of economic minerals from the open pit and underground mines. These waste rocks are directly dumped on the ground surface (Wang et al., 2013). The surface disposal of waste rocks has both social and environmental impacts. The rock filling is applicable only in some cases of underground mining. It facilitates the increase in waste rock utilization, prevents loss of valuable land, minimizes adverse environmental effects of surface disposal, improves the stability of mine workings, and prevents subsidence. Waste rocks used for backfilling purpose, which has not undergone particle grading, can be classified as unmodified rock fill or simply rock fill (Potvin et al., 2005). Rock fill is a type of uncemented backfill mostly consisting of waste rocks produced during mine development. Rock filling may be defined as the backfilling method where the backfill materials such as industrial solid wastes, soil, stone, boulders, gravel, etc., are transported to underground stopes for filling the mine voids. In this method, neither water nor any binder is used (Yao et al., 2012).

The rock fill is composed of rock fines and gravel to boulder size materials. Rock fines consist of rock particles less than 10 mm, whereas, gravel size ranges between 10 to 60 mm and boulder consists of rocks greater than 150 mm size (Yao et al., 2012).

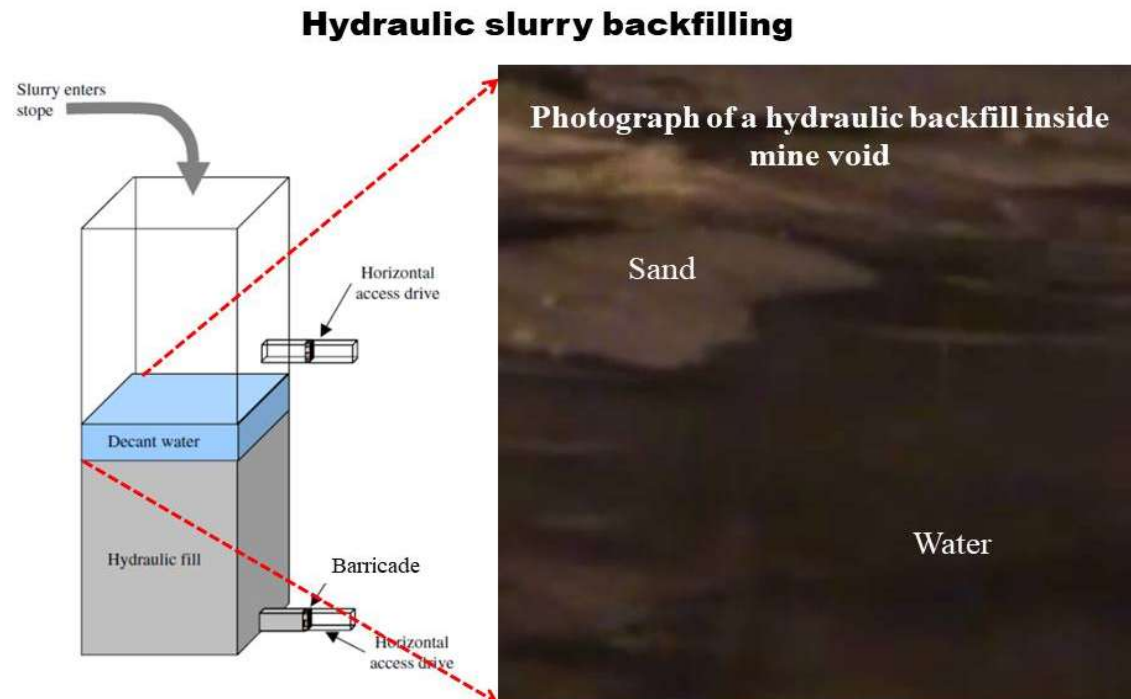
Rock Filling



METHODS OF BACKFILLING IN METALLIFEROUS MINES

Hydraulic Slurry Backfilling Solid: 40-50% by weight, Water

Hydraulic slurry backfilling in underground mines is one of the uncemented backfilling methods where water is used as the conveying medium to transport the backfill materials. The hydraulic fill generally consists of 40-50 wt% solid with less than 10% particles below 50 μm size. The backfill materials can be mill tailings, slag, sand, and crushed rock. Before the placement of hydro-fill, a few prior arrangements, such as barricade construction and laying of drainage or decant pipes, are made at the draw or decant level. During hydro-fill pouring into the stopes, the height of the backfill material increases and the water accumulates on the upper portion of the backfill. This accumulated water needs to be drained out through the barricades as seepage or decanted water (Potvin et al., 2005; Zheng and Li, 2020).



METHODS OF BACKFILLING IN METALLIFEROUS MINES

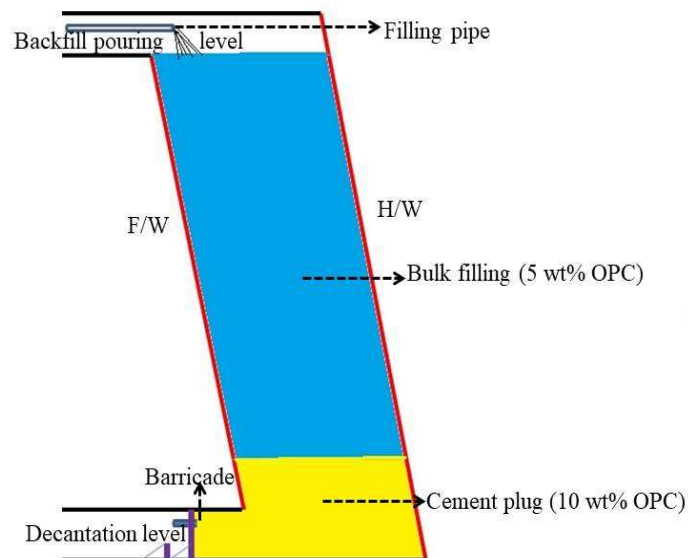
Cemented Hydraulic Slurry Backfilling Solid: <60% by weight, Binder: (OPC), Water

The conventional hydraulic backfilling can be converted to cemented hydraulic slurry backfilling (CHSB) by adding a suitable binder. Typically, ordinary Portland cement (OPC) is used as binder in cemented hydraulic backfilling. The addition of OPC increases the strength and cohesion of the backfill, thereby making the backfill self-standing after the extraction of adjacent stopes. In hydraulic backfilling, the backfill is hydraulically transported to underground stopes through pipes. The pouring is strategically done in the upper end of the stopes to achieve the best backfill results (Liston, 2014). The cemented hydraulic slurry backfill typically consists of classified tailings or sand or processed overburden (OB) with cement as a binder, and the solid concentration in the slurry is below 60 wt%. The efficacy and economy of backfilling are dependent upon the cement content and slurry density (Masniyom, 2009). Generally, the OPC content in the CHSB is relatively higher at the decantation level, termed as plug, than the main hydro fill (i.e., above the fill plug).

High Concentration Filling: Solid: 60 - 70% by weight

High concentration filling (HCF) is an advanced version of the CHSB, which uses cemented hydraulic backfill with a solid content varying between 60 to 70 wt%. The high density of backfill minimizes the water drainage. The target backfill slurry density is a function of solid wt% and particle specific gravity. It balances the requirements of transporting the slurry without plugging the pipeline and excess drainage of water. HCF is placed in underground either by gravity or using pumps depending on the site requirement (Pal et al., 2009). The excess ponded water inside the stopes is drained out through decantation pipes or suitable drainage system (Sainsbury, 2003).

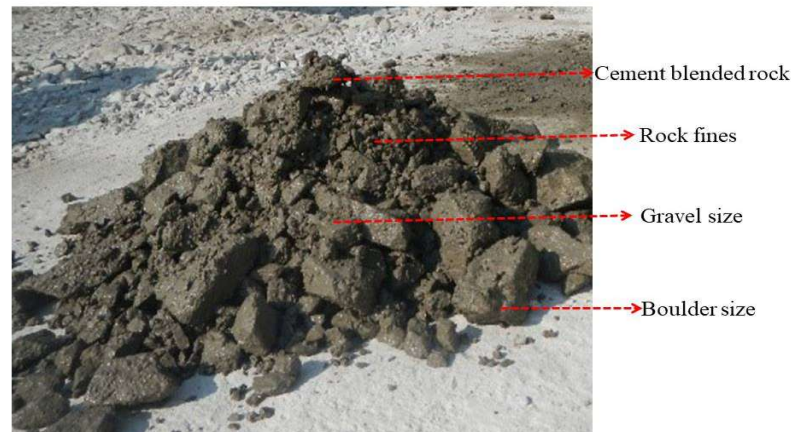
Cemented hydraulic slurry backfilling



Cemented Rock Filling Solid: Screened to remove large boulders & Cement is added

Rock fill is loosely packed and unable to stand in vertical exposure. It only serves as a support to the stope hang wall. Therefore, it is not possible to mine next to the rock-filled stopes. The rock fill can be transformed into cemented rock fill (CRF) by mechanically adding cement to it. Hence, CRF consists of unclassified waste rock mixed with cement slurry to provide bond strength between the interlocking rock particles (Sainsbury, 2003). In a typical CRF, unclassified waste rock having large boulders (150 mm) to fines (< 10 mm). It is a kind of unmodified rock fill that can be modified to be self-supporting during vertical exposure and allow mining next to the fill by reducing the porosity of placed backfill. It is done by adding binders and optimization of the particle size distribution (PSD) of the final backfill mixture (Potvin et al., 2005). Generally, in CRF, the waste rock generated during mine development is screened to remove the large boulders before mixing. Cement can be mixed with the waste rock in batches by blending the constituents with a load haul dumper (LHD) in underground. Cemented rock fill typically has high strength in the range of 4-7 MPa and used for weak rock mass (Pakalnis et al., 2007). In addition, it has a higher stiffness in comparison with paste backfill (Rai et al., 2013). High strength backfill material is required to prevent higher rate of wall closure in the weak rock mass and minimize ground movement (Hughes, 2014; Grobler et al., 2019). In practice, only few mine operators can afford to crush and screen the rock fill to an ideal size. However, Stone (1993) suggested that the blending of two or three size ranges can significantly improve backfill strength.

Cemented rock fill (CRF)



Cemented Aggregate Filling

Cemented backfill composed of engineered rock fill or aggregate, termed as cemented aggregate fill (CAF). Large-scale cemented aggregate fill operations in Kidd Creek and Williams Mines in Ontario, Canada (Yu, 1989; Durkac et al., 2001) utilized a system of fill raises and conveyors to transport and place the cemented aggregate fill. Bloss and Greenwood (1998) reported that the segregation of rock fill particles could not be avoided due to the mechanism of backfill placement. During the placement of CAF, a zone of fine aggregates occurs near the impact area, consuming most of the cement and leaving a low cement content rock fill at the perimeter of the backfilled cone (Sainsbury, 2003). Yu (1989) observed that segregation of rock fill is more pronounced when the conveyor fills the stopes. The segregation of cemented rock fill and aggregate causes significant variability in the behaviour and failure mechanism after subsequent exposures of the fill material.

Cemented aggregate fill (CAF)

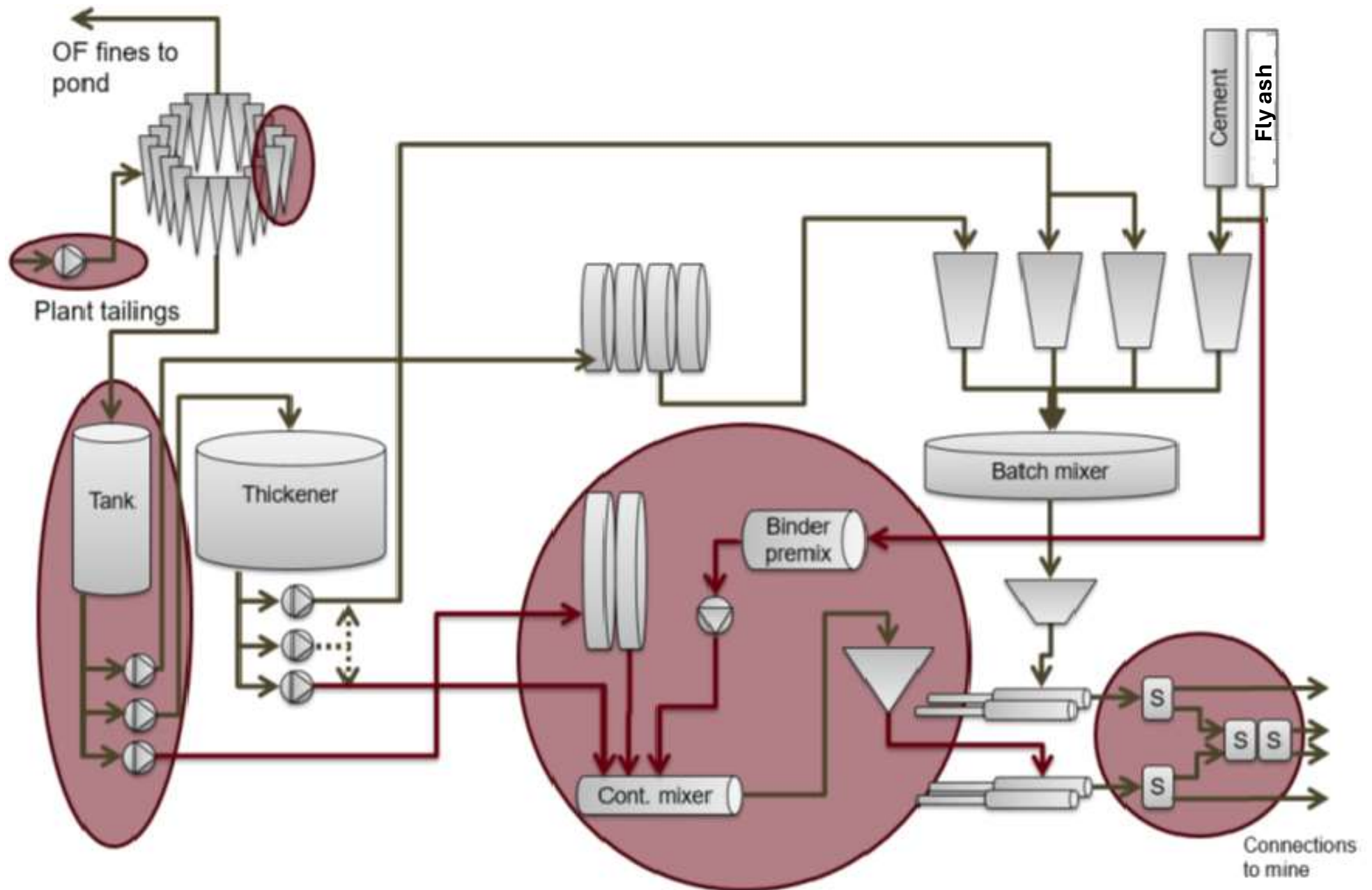


Paste Backfilling Solid: 70 - 80% by weight, Binder: 1 - 12% by weight

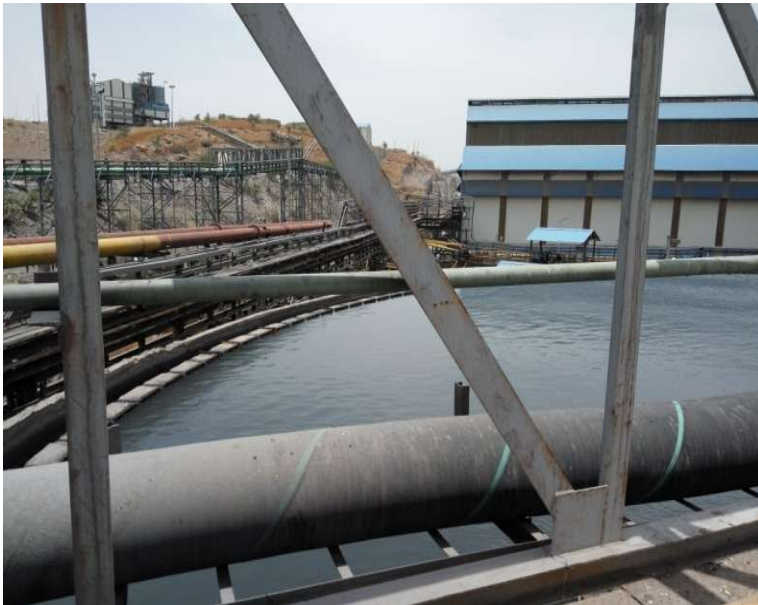
Paste backfilling is the advanced version of cemented backfilling, in which the mill tailings are mixed with a definite dosage of binder, such as OPC, fly ash, lime, and slag (Sivakugan et al., 2006b). The optimum solid to water ratio in paste backfilling is maintained in such a way that the backfill mix is easily flowable. As compared to other backfilling techniques, paste backfilling consists of a higher solid concentration (70-80 wt%). Potvin (2005) defined paste as a mixture of material, which akin to toothpaste, comprising 75% to 85% solid by weight and 1 wt% to 12 wt% binder, containing at least 15 wt% solid fractions passing 20 μm . When placed, it does not bleed water, does not settle or segregate in the pipeline, and has slump less than 250 mm. Niroshan (2018) reported that paste backfill usage results in a reduction of barricade preparation requirements, high rate of solid disposal in stopes, and, therefore, an overall reduction in mining cycle time by 50%.

The paste backfill is prepared from classified mill tailings with the addition of a suitable binder and water. The prepared paste backfill is transported into the stopes under gravity or by using positive displacement pumps (Brackebusch, 1994). A reticulated pipeline system is used for the transportation of paste backfill from the surface to underground voids. It provides a flexible system for coping with complex mining structures. The use of underground paste backfill enhances ground support, minimizes surface subsidence, roof fall, and improves ore recovery and productivity (Deb et al., 2017; Wu et al., 2020). It also prevents possible liquefaction because of the lower water content in the paste backfill (Qi et al., 2019d). Hence, it generates comparatively lesser pressure on the backfill barricades (Li and Aubertin, 2009a, 2009b).

Layout of paste fill plant



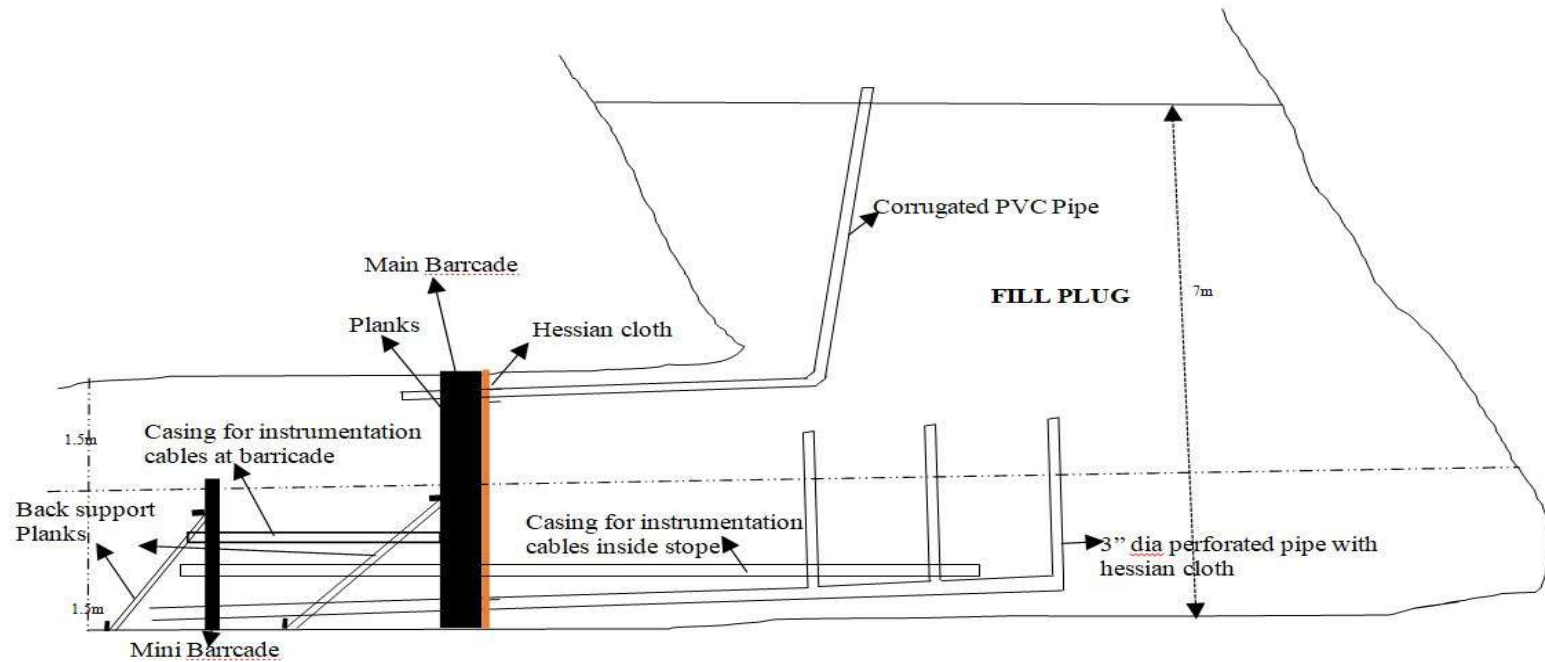
Paste Fill Plant



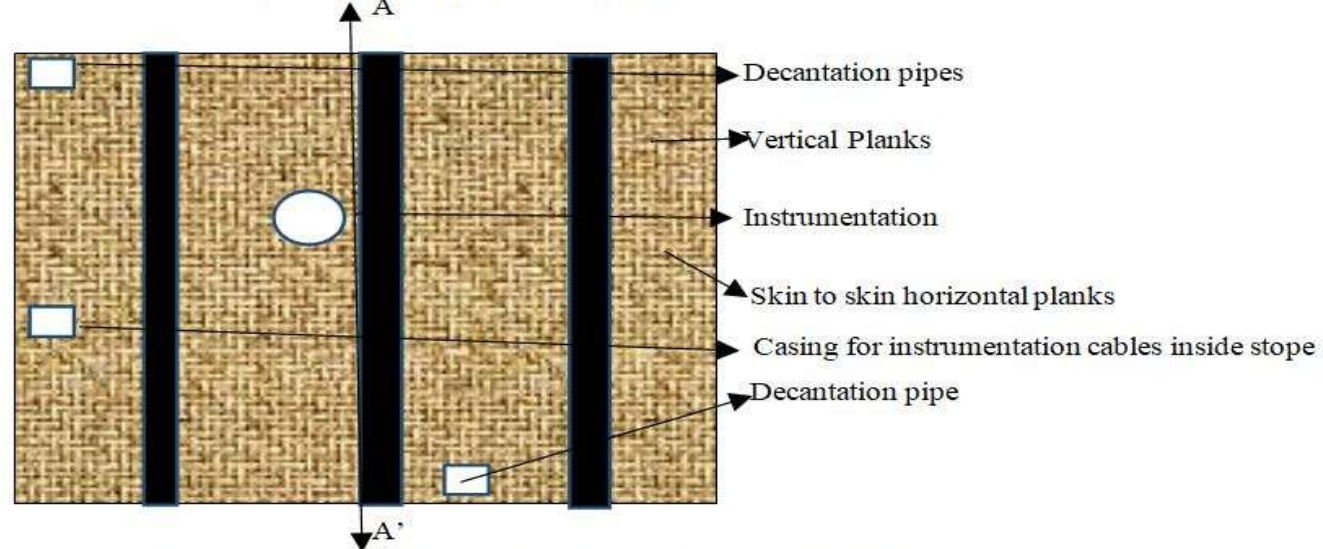
Benefits of efficient barricade

- Economic benefits:
 - » increases in fill pour rate
 - » continuous paste pours
 - » widening of stope dimensions dictated by barricade loads.
- Safety benefits:
 - » discontinuities between successive paste pours would be minimized
 - » the risk of designing a backfill stope that would exceed a barricade's capacity is largely eliminated.

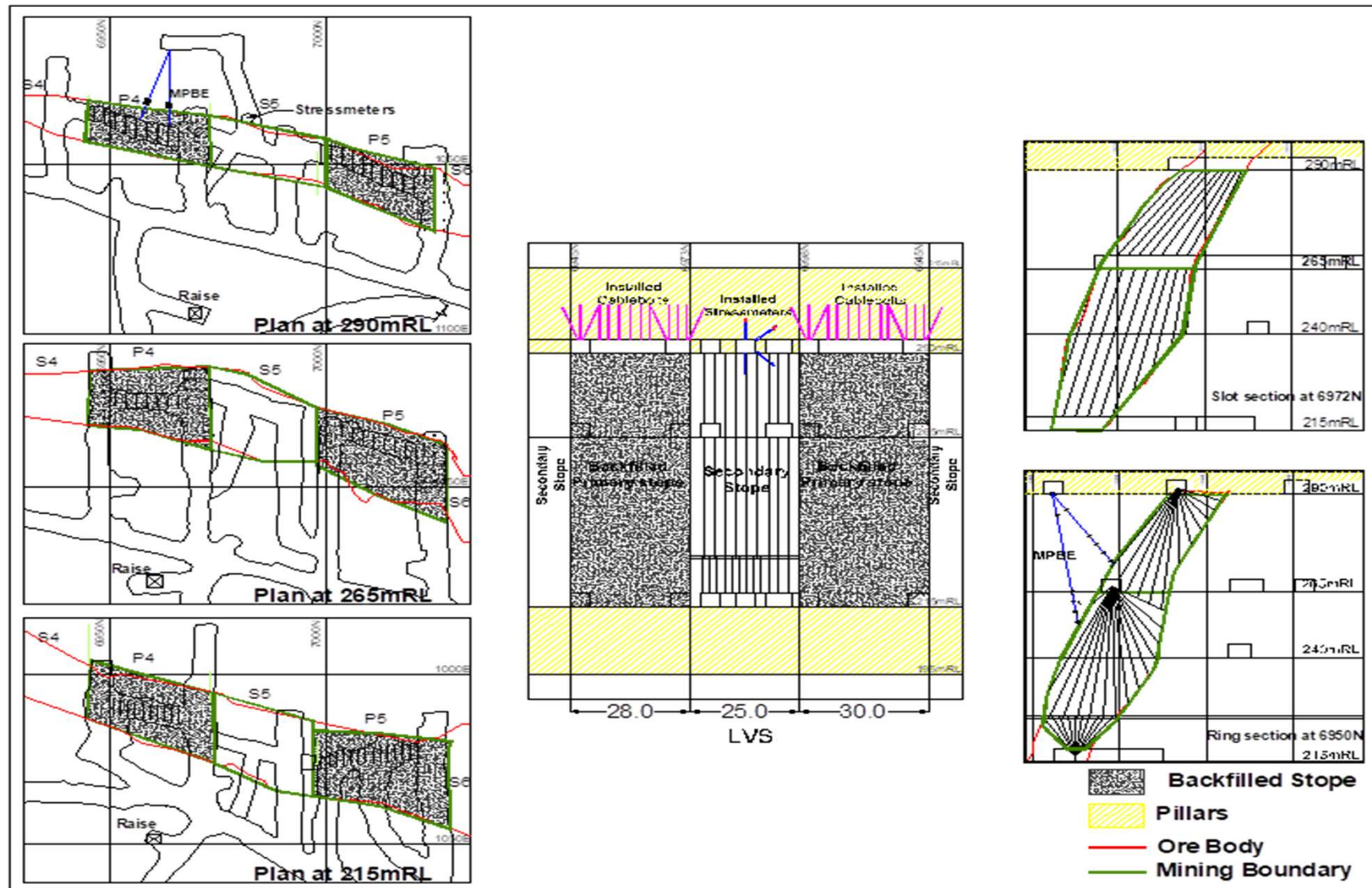
Barricade construction



SECTION ALONG A-A'



Primary secondary mode



Primary secondary mode

